PDF DUO

User guide — PDF, electrical diagrams, GRAFCET and floor plans — free software

Version 1.0.0 · Jacques Branas, en collaboration avec Claude.ai

PDF DUO lets you read, fill in and sign PDFs, draw electrical, pneumatic and network diagrams, GRAFCETs and floor plans, and convert Word / ODF ↔ PDF. It is designed for vocational-school students and teachers, and is also handy for school life: posters, membership forms and signed documents for the student association. The red numbers on the screenshots refer to the lists below them.

1. Installation

Windows: run the installer `PDF-DUO-Windows-Setup.exe`, or unzip `PDF-DUO-Windows.zip` and open `PDF DUO.exe`.

Linux: double-click `pdf-duo_amd64.deb` (Debian, Ubuntu, Mint), or use the AppImage (make it executable, then open it).

macOS: pick the Apple Silicon (M1, M2…) or Intel version, unzip, drag “PDF DUO” to Applications. First launch: right-click › Open.

Word → PDF conversions: PDF DUO uses **LibreOffice** if installed (most faithful, free), otherwise **Microsoft Word**, otherwise its own converter (simplified layout). ODT files need LibreOffice.

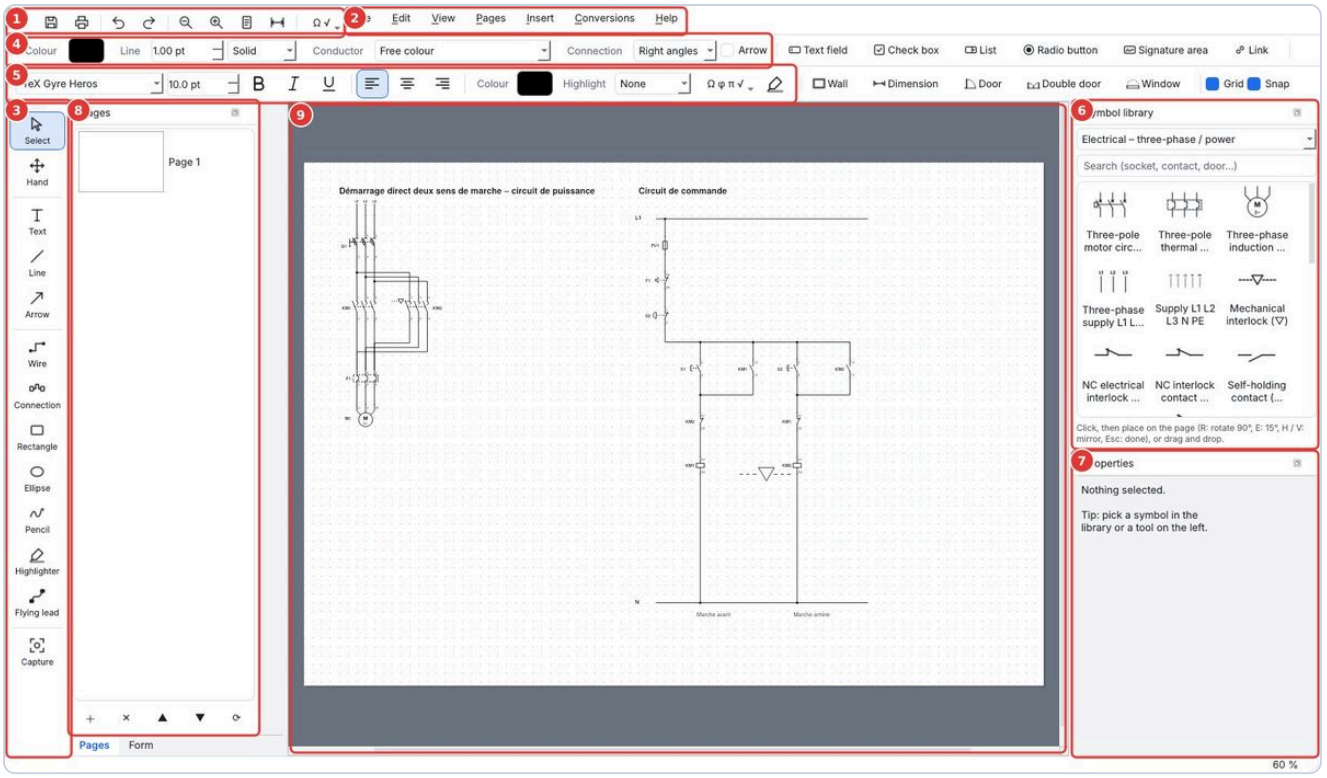
2. The window

Top, left of the menus: open, save, print, **undo ↶ / redo ↷**, **zoom − / +**, whole page, special characters **Ω √**.

Left: the tools (select, text, line, wire, connection, shapes, pencil, highlighter, flying lead, capture).

Right: the **symbol library** (choose a category or type a word to search) and the **Properties** panel of the selected item.

View › *Palettes* shows or hides each panel; *View* › *Theme* and *Language* change the look and language.



The PDF DUO window

1. Quick bar: open, save, print, undo, redo, zoom − / +, Ω √ characters
2. Menus
3. Drawing tools
4. Colour, width, conductor, connection type, arrow
5. Writing palette: font, size, bold, colour, highlighter
6. Symbol library (categories and search)
7. Properties of the selected item
8. Pages (thumbnails) and Form tab
9. The working page

3. Opening and saving

File › Open accepts PDFs, Word / ODT documents (converted automatically) and images.

Drawings are saved as vectors in the PDF. When the file is reopened in PDF DUO, everything stays editable.

File › Export a final PDF produces a flattened PDF to hand out.

4. Drawing a diagram

Click a symbol in the library, then click on the page to place it (or drag and drop). **Esc** to finish.

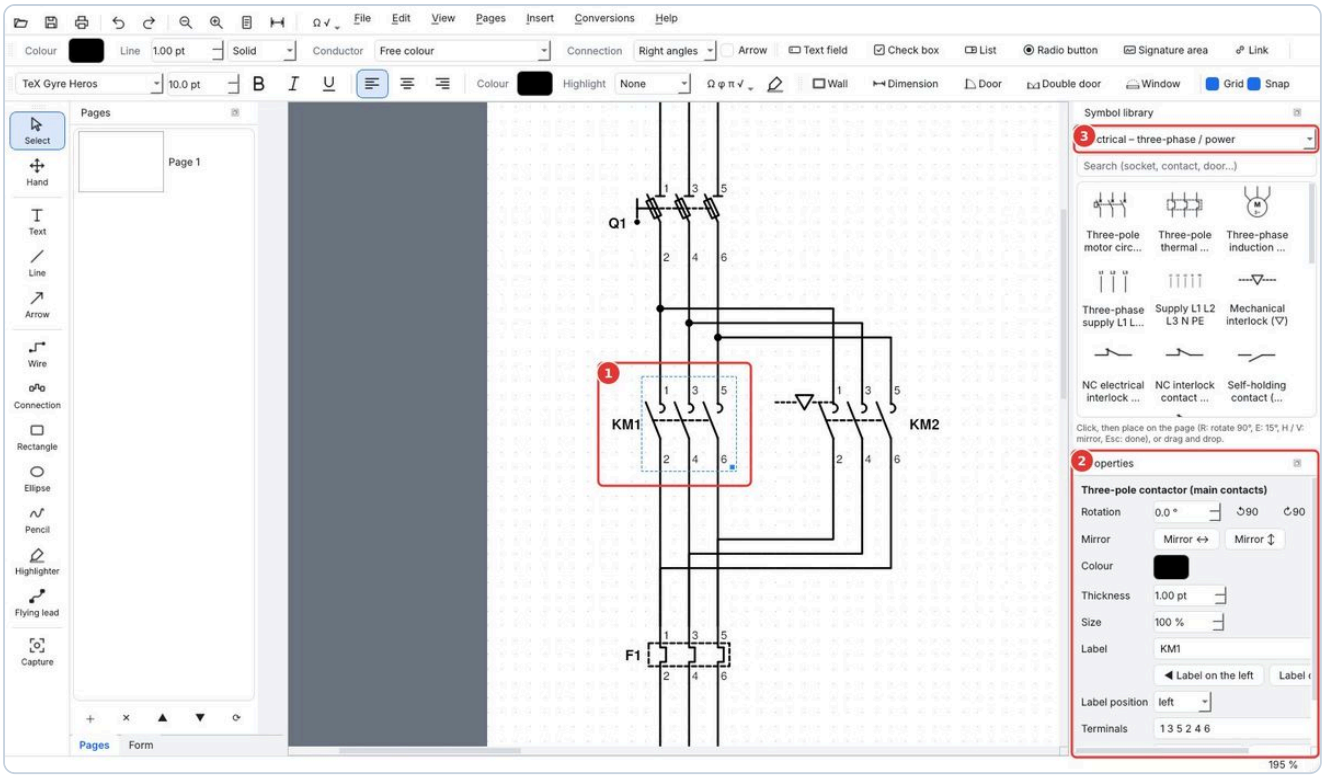
R rotates 90°, **E** 15° (Shift = other way), **H** / **V** mirror. Exact angle in Properties.

The **blue handle** at the bottom right enlarges the item proportionally.

The **label** (KM1, S1…) and **terminal numbers** are edited in Properties; “Label on the left / right” buttons.

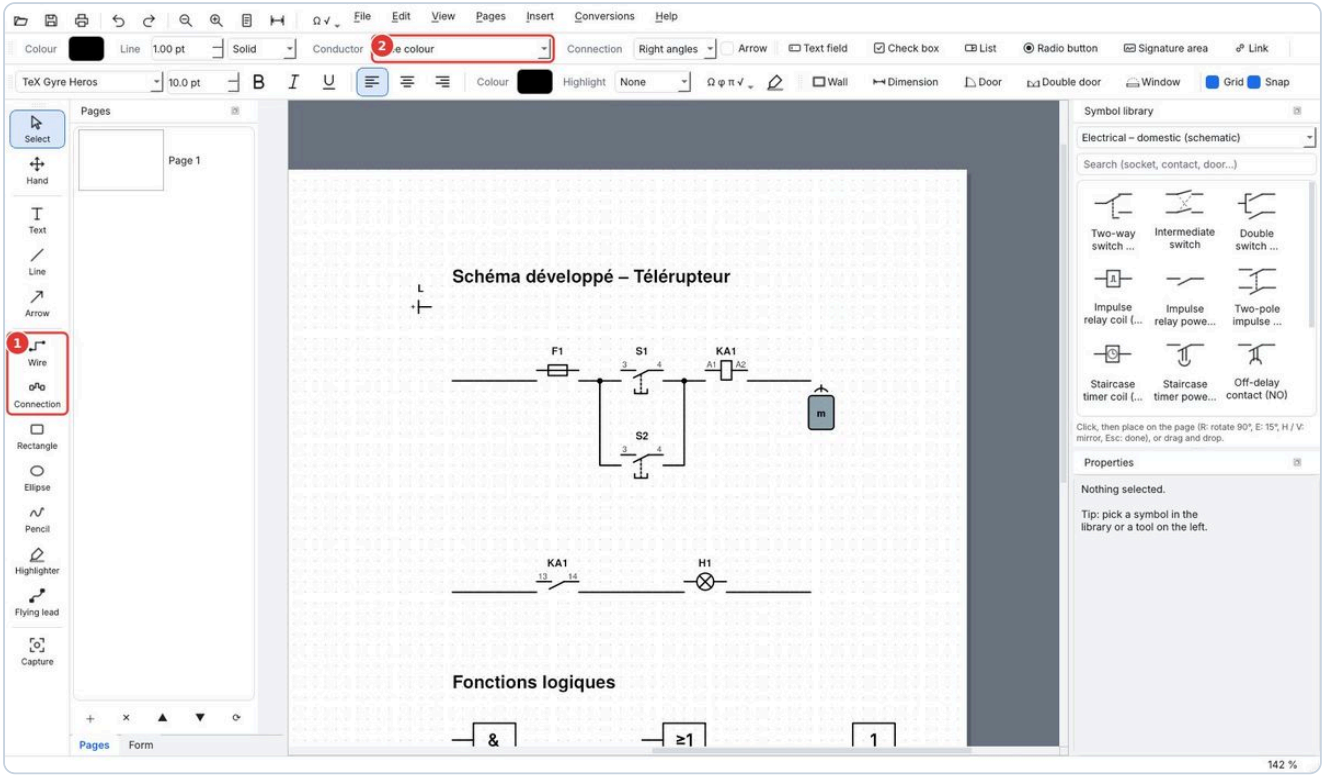
Wire tool: one click per corner; clicking a terminal (small circle) attaches the wire. **Connection** tool: click two terminals and the route is drawn. Wires follow moved symbols.

Conductor list: standard colours (phase, blue neutral, green/yellow earth, fibre, Ethernet…). Junction dots are placed automatically.



A selected symbol and its properties

1. Selected symbol (dotted frame): blue handle at the bottom right to enlarge it
2. Rotation, mirror, colour, width, size, KM1 label and its position
3. Choice of the symbol category



Wires, connections and conductor colours

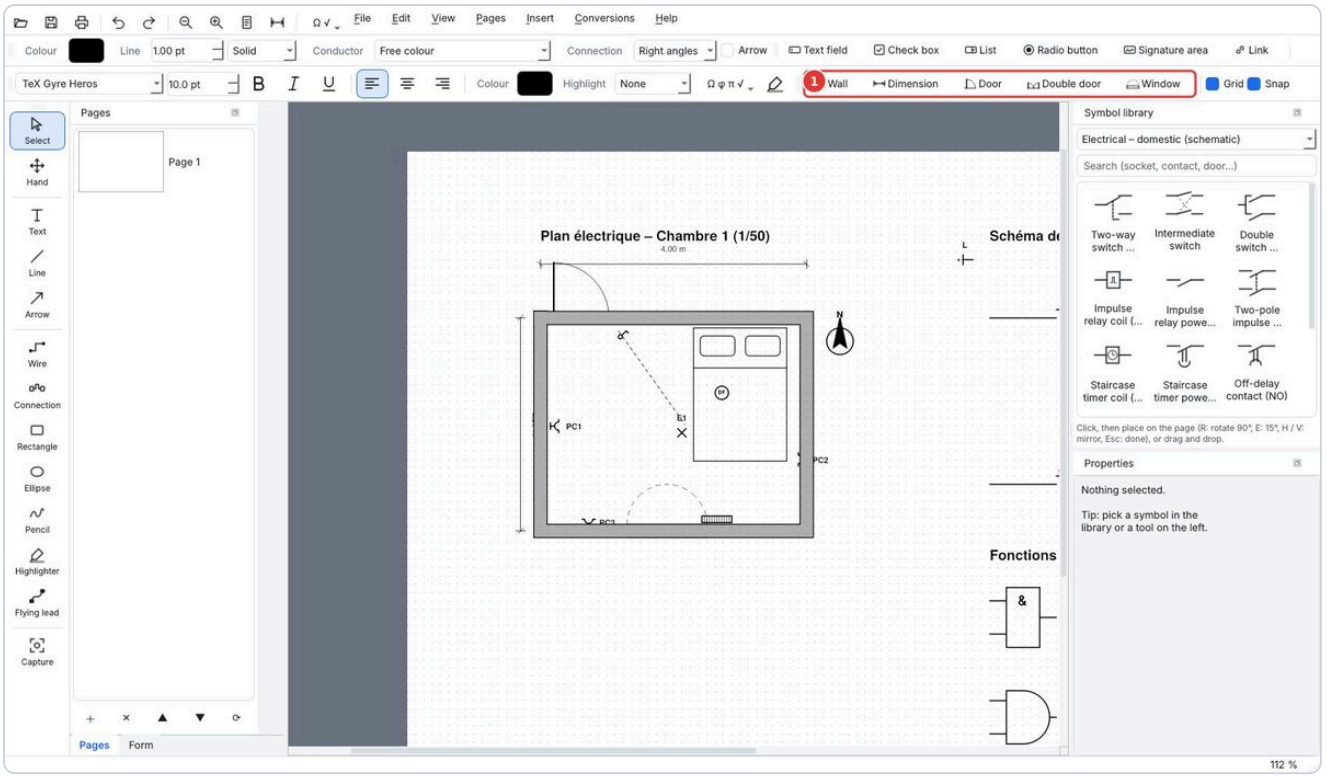
1. Wire and Connection tools
2. Standard conductor colour

5. Floor plans

Set the **scale** (1/50, 1/100…) and the real grid step in the *Plan* toolbar.

Wall tool: one click per corner, double-click to finish. **Door**, **Double door**, **Window** tools: move over a wall and the opening aligns and takes the wall thickness.

Dimension tool: drag from point to point, the real length is shown. *Insert › Title block* adds the title block.



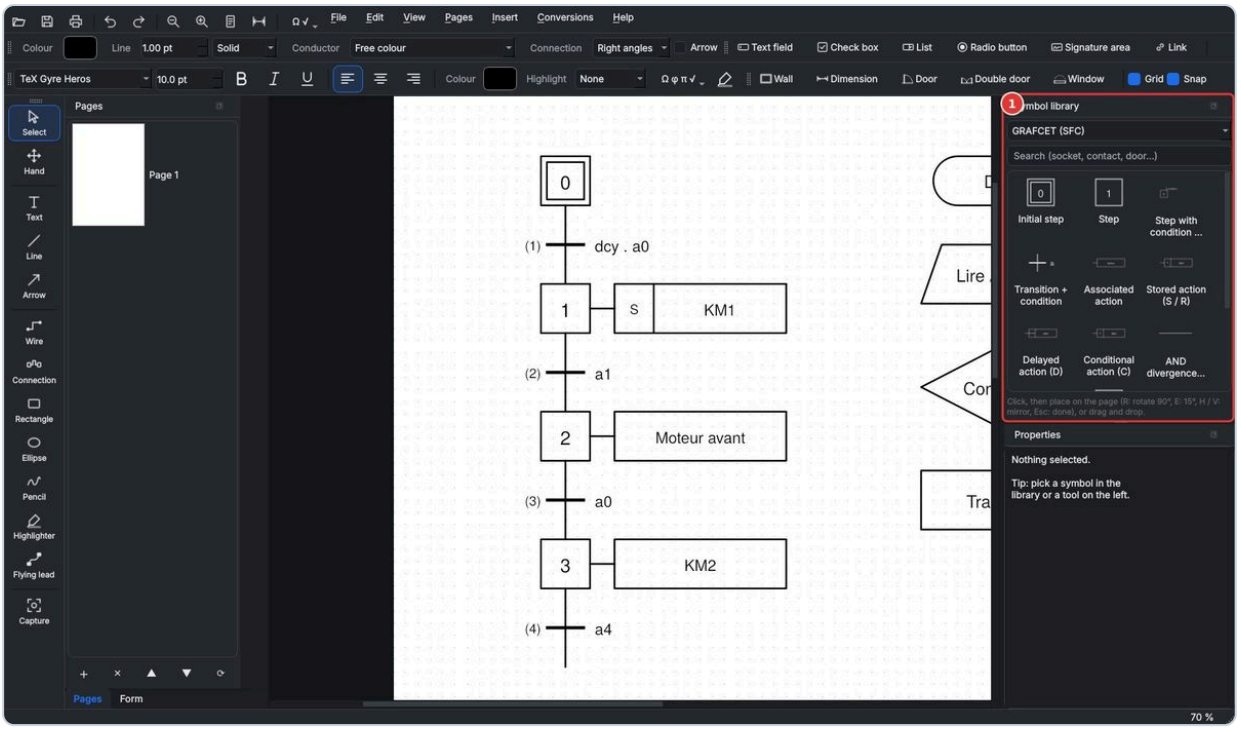
Plan of a bedroom at 1:50: walls, door, window, sockets, dimensions

1. Wall, Dimension, Door, Double door, Window tools

6. GRAFCET, flowcharts, mind maps

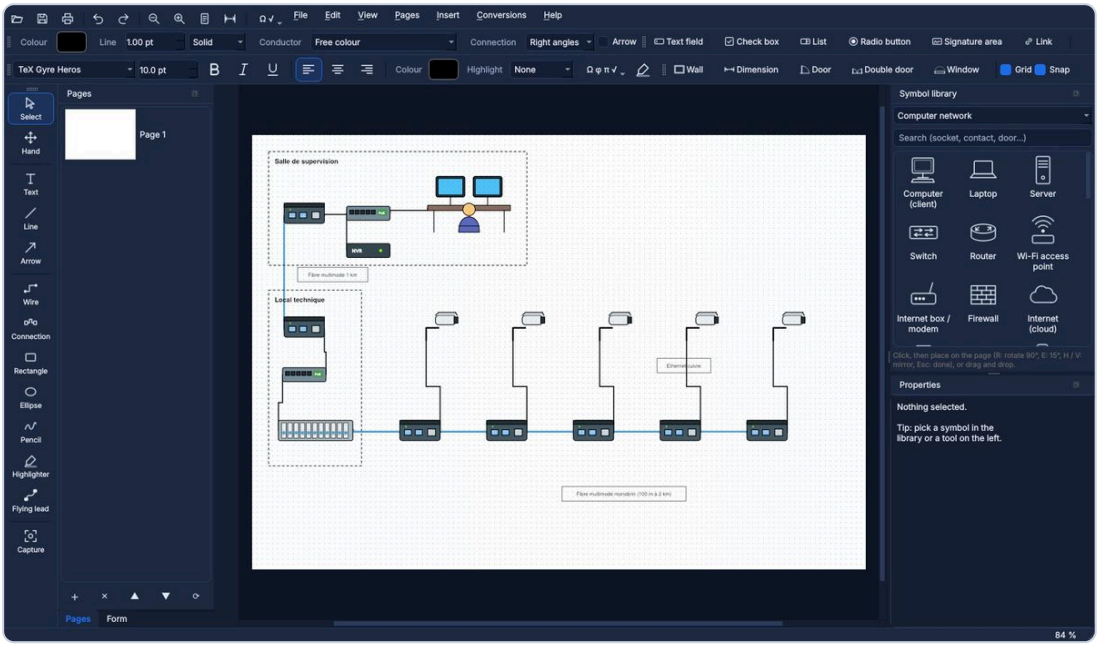
Place an initial step, select it, then press **Ctrl+G**: a transition and the next step are added, connected and numbered.

Double-click a shape to type its text. Choose *Connection: curved* for mind maps, tick *Arrow* for flowcharts.



A GRAFCET: steps, transitions, actions

1 GRAFCET category of the library

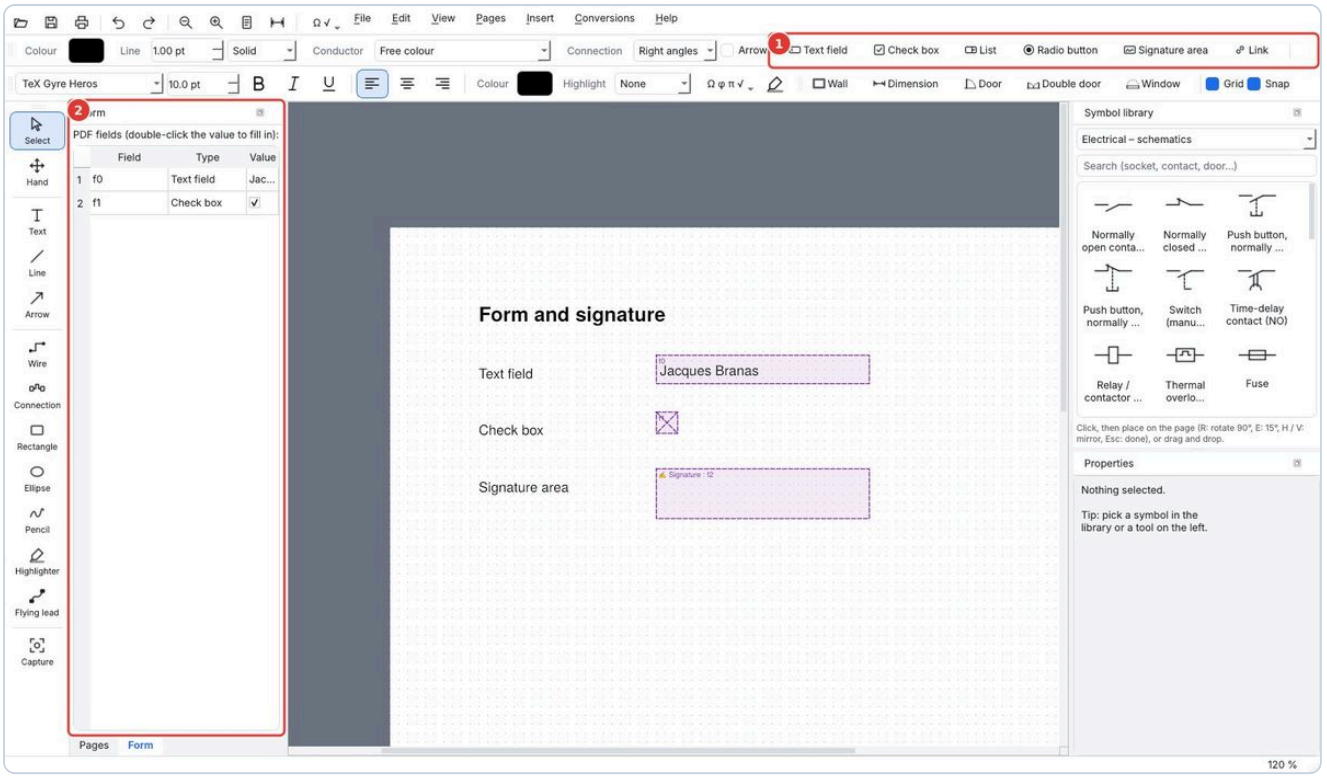


CIEL network diagram (CCTV, fibre) — midnight blue theme

7. Forms and signature

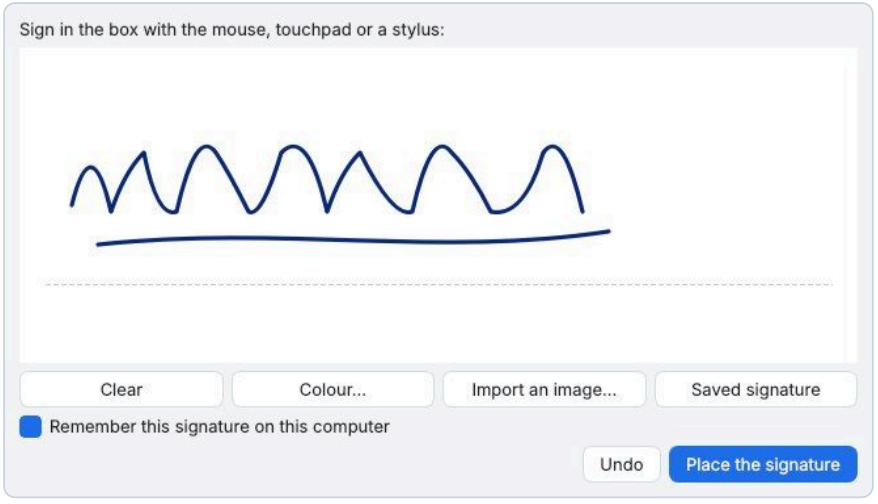
Form and signature toolbar: text field, check box, list, radio button, signature area, link. Double-click to fill in; the *Form* panel lists everything.

Signature button: draw or import your signature, then click where to place it.



Fillable form fields

1. Form and signature toolbar
2. Form panel: all fields and their values

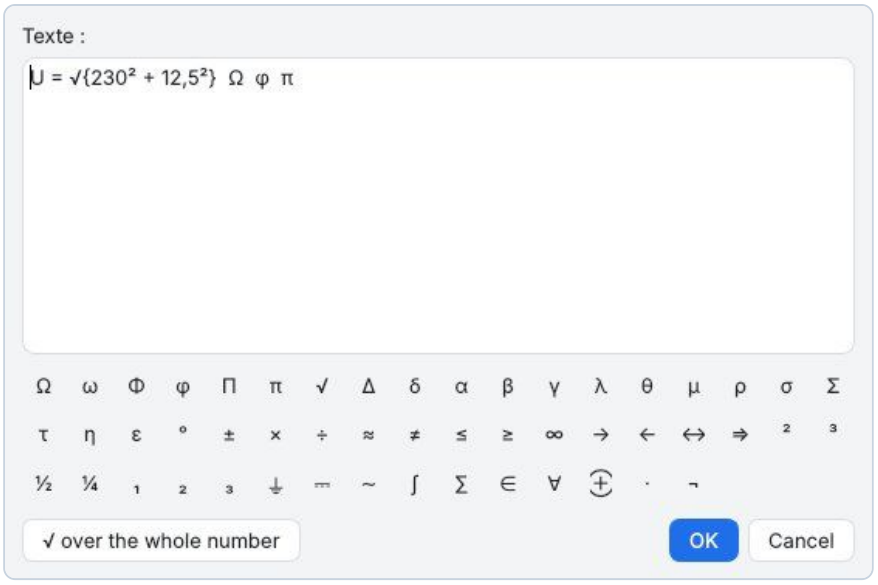


The Signature window

8. Text and special characters

Writing palette: font, size, bold, italic, underline, alignment, colour, highlight.

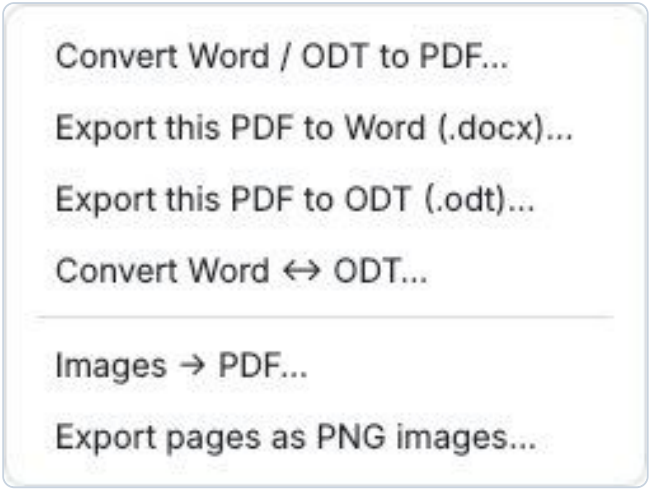
Ω √ button: Ω, φ, π, Δ, μ, °, ±… Type `√{12.5}` : the root bar covers the whole number.



Text with special characters and a root over the whole number

9. Conversions

Conversions menu: Word / ODT → PDF, PDF → Word, PDF → ODT, Word ↔ ODT, images → PDF, pages → PNG.

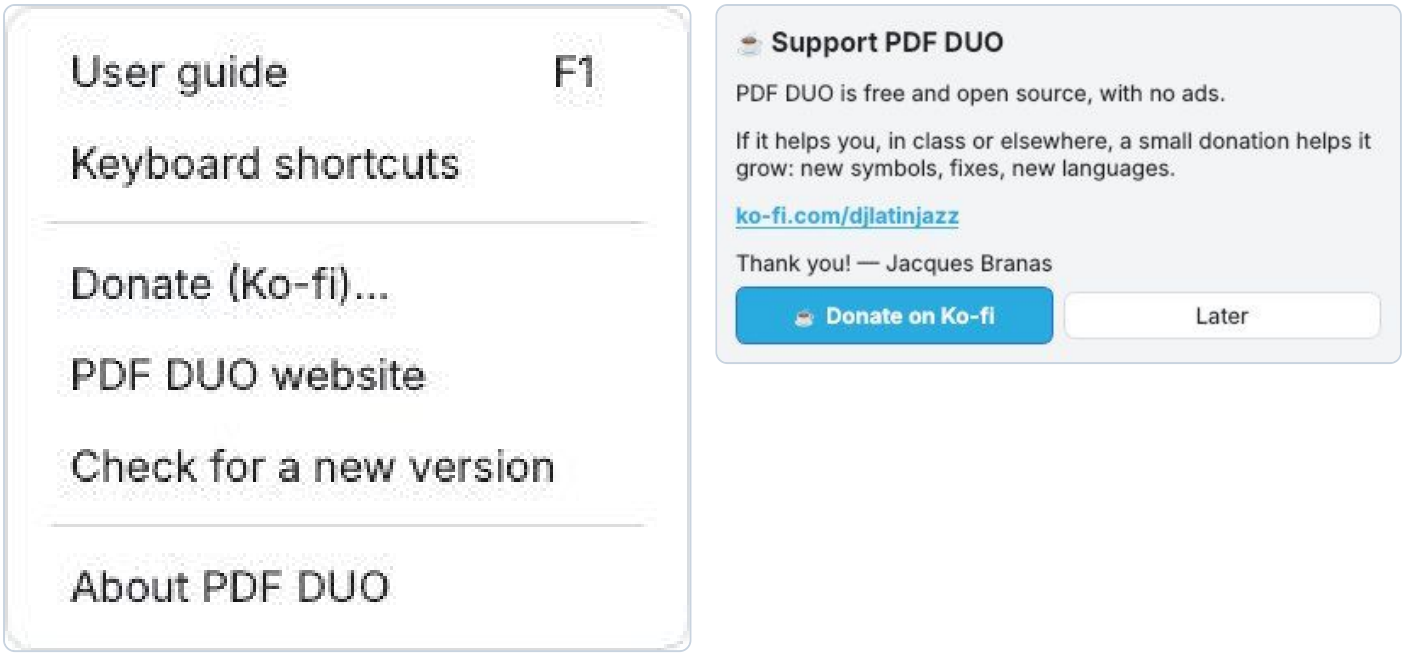


The Conversions menu

10. Support the project

PDF DUO is free software (GNU GPL v3). A donation at **ko-fi.com/djlatinjazz** helps it grow (*Help › Donate*).

Website: deltapassion.github.io/PDF-DUO — Contact: deltapassion@gmx.com



Help menu and donation window